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Chapter 1

Sampling and Monte Carlo Methods

1.1 Motivation: Numerical Integration

The goal is to **generate random variables from a distribution** and to **find expectations** via integration.

In general we need to compute

$$\mathbb{E}[g(X)] = \int_{\mathcal{X}} g(x) f_X(x) dx,$$

where f_X is the density of X .

Remark

Frequentist inference. To obtain the variance of an estimator one needs the Fisher information, i.e. the expectation of the second derivative of the log-likelihood.

Bayesian inference. One needs $\mathbb{E}[\theta]$ under the posterior distribution.

There are ways of doing integration *numerically*. Suppose

$$g : D \rightarrow \mathbb{R}, \quad D \subseteq \mathbb{R}^n, \quad I = \int_D g(x) dx.$$

(For $n = 1$ this is simply $\int_a^b g(x) dx$.)

If $n = 1$, **quadrature methods** can be used to approximate the integral. If m is the number of sub-intervals the error rate is $O(m^{-k})$, where k depends on the method used.

Example: Simpson's rule

If Simpson's method is used, then k is the number of nodes.

1.1.1 Drawbacks of Numerical Integration

- (1) We know the *rate* but do not have actual bounds on the error:

$$\hat{I} - I = O(m^{-k}).$$

We do not know the constant in $O(\cdot)$.

- (2) For n dimensions the order is $m^{-k/n}$. We need m^n function evaluations to get the same level of accuracy a grid with m points per dimension requires m^n total points. This is the *curse of dimensionality*.

1.2 Monte Carlo / Simulation Methods / Sampling

Sampling / Simulation / Monte Carlo methods overcome these drawbacks.

Key idea (Law of Large Numbers). Generate random variables

$$X_1, \dots, X_m \stackrel{\text{i.i.d.}}{\sim} U(D).$$

(1) **(LLN)** The sample mean

$$\bar{g}_m(x) = \frac{1}{m} \sum_{i=1}^m g(X_i) \xrightarrow{m \rightarrow \infty} \frac{\int_D g(x) dx}{\text{area}(D)}.$$

(2) **(CLT)** Normalised fluctuations are asymptotically normal:

$$\sqrt{m} \frac{\bar{g}_m(x) - \frac{\int g(x) dx}{\text{area } D}}{\sqrt{\text{Var}(g(X))}} \Rightarrow N(0, 1).$$

This result *does not depend on the dimension of D* (area is replaced by volume of D in higher dimensions).

The estimator of I and its error are:

$$\hat{I}_1 = \bar{g}(x) \cdot \text{area}(D), \quad \hat{I}_1 - I = \frac{\sqrt{\text{Var}}}{\sqrt{m}} \cdot \text{area}(D) \cdot Z, \quad Z \sim N(0, 1).$$

We therefore have a good idea of the bound on the error (it is a normal random variable).

Example: Estimating $\int_0^1 \cos x dx$

Let $g(z) = \cos z$, $D = [0, 1]$. Then $I = \sin 1 = 0.841471$.

m	$\hat{I} = \bar{g}(X)$
1	0.774
10	0.872
100	0.834
10^4	0.840
10^5	0.841

1.3 Generating Uniform(0,1) Random Variables

We need a sequence $U(0, 1)$ values. The built-in function in R is `runif`.

Requirements for a good generator:

1. Each observation should be $U(0, 1)$.
2. Independent.

1.3.1 Pseudo-Random Numbers

Real / physical random numbers can be generated by physical processes (e.g. coin flips). In practice we use **pseudo-random numbers**.

Linear Congruential Generator

Start with a seed $z_0 \rightarrow$ integer. Fix integers a and m :

$$z_1 = az_0 \pmod{m}, \quad z_2 = az_1 \pmod{m}, \quad \dots$$

so $z_i \in \{0, 1, \dots, m-1, m\}$ and

$$U_i = \frac{z_i}{m}.$$

The sequence $U_1, U_2, \dots$ is the output.

Note

m should be *large* (avoid $z_i = 0$ or m being prime). If $a = 2$, the cycle has length 8. If you take too many random variables there will be repetition. There is clear **dependence** between (U_i, U_{i+1}) , $i = 1, \dots, n$, ex. a *lattice structure*.

1.3.2 Combining Generators

Use several linear congruential generators with parameters $(a_1, m_1), \dots, (a_k, m_k)$, producing sequences $z_{1j}, \dots, z_{kj}$ and $U_{1j}, \dots, U_{kj}$. Combine:

$$Z_i = \sum_{l=1}^k (-1)^l z_{li}, \quad m = \max(m_1, \dots, m_k),$$

$$U_i = \begin{cases} Z_i/m & \text{if } Z_i > 0, \\ \text{no output} & \text{otherwise.} \end{cases}$$

where $U_1 = \left\{ \sum_{l=1}^k U_{l1} \right\}$ (fractional part).

1.4 Generating General (Non-Uniform) Random Variables

Many standard distributions have built-in R functions to generate them: `rbinom`, `rexp`, `...`

If you want to generate from a **non-standard distribution**, one general procedure is to use the **Inverse CDF** method.

1.4.1 Inverse Transform Method

Let $X \sim F_X$, so $P(X \leq x) = F_X(x)$. Define the generalised inverse:

$$F_X^{-1}(u) = \inf\{t : F_X(t) \geq u\}, \quad 0 \leq u \leq 1.$$

Theorem: Inverse Transform

$F_X(x) \geq u$ if and only if $F^{-1}(u) \leq x$. —(i)

If $U \sim \text{Unif}(0, 1)$, set $X = F^{-1}(U)$. Then

$$P(X \leq x) = P(F^{-1}(U) \leq x) = P(F(x) \geq U) = F(x),$$

using (i) and the fact that $U \sim \text{Unif}$.

Example: Exponential distribution

$X \sim \text{Exp}(1)$: $F(x) = 1 - e^{-x}$. Setting $F(x) = u$:

$$F^{-1}(u) = -\ln(1 - u).$$

1.4.2 Kolmogorov–Smirnov (KS) Test Connection

Given $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$, the KS statistic is

$$\sup_x |F(x) - \hat{F}_n(x)|.$$

To compare two samples $X_1, \dots, X_n \sim \text{Exp } F_X$ (using built-in `rexp`) and $Y_1, \dots, Y_n \sim \text{Exp } F_Y$ (using inverse CDF):

$$\sup_{u \leq t \leq 1} |\hat{F}_X(t) - \hat{F}_Y(t)|.$$

1.4.3 Inversion Directly

Logistic distribution.

$$F(x) = \frac{1}{1 + \exp\left(-\frac{x - \mu}{\lambda}\right)} = u \implies F^{-1}(u) = -\lambda \ln\left(\frac{1}{u} - 1\right) + \mu.$$

Cauchy distribution.

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan\left(\frac{x - \mu}{\sigma}\right).$$

1.5 Generating from Distributions Using Properties

Use exponential and properties of random variables / distributions.

- Gamma(α, β) where α is an integer is the sum of α many independent exponentials.
- $\chi_{2n}^2 = \text{Gamma}(n, \frac{1}{2})$.

- $\text{Beta}(n_1, n_2) = \frac{X_1}{X_1 + X_2}$, where n, n_1, n_2 are all integers, $X_1 \sim \text{Gamma}(n_1, 1)$, $X_2 \sim \text{Gamma}(n_2, 1)$ independently.

We need other procedures, as we cannot write the CDF in closed form or invert it analytically for many distributions.

1.5.1 Discrete Distributions

Let

$$X = \begin{cases} x_1 & \text{with prob. } p_1 \\ \vdots & \\ x_k & \text{with prob. } p_k \end{cases}, \quad \sum_{i=1}^k p_i = 1, p_i > 0, x_1 \leq x_2 \leq \dots \leq x_k.$$

Let $U \sim \text{Unif}$. Compute $F(x_i) = \sum_{j=1}^i p_j$. Find i such that

$$F(x_{i-1}) < u \leq F(x_i), \quad \text{then set } X = x_i.$$

1.5.2 Multinomial Distribution

Let $\mathbf{X} = (X_1, \dots, X_k)^T$ with $\mathbf{X} \sim \text{Mult}(n, \mathbf{p})$, $\mathbf{p} = (p_1, \dots, p_k)^T$, $p_i \geq 0$, $\sum p_i = 1$, $x_i \geq 0$, $\sum x_i = n$.

n independent trials, each trial can lead to k outcomes; the j -th outcome happens with probability p_j .

Remark

Inverse transform: $U \sim U(0, 1)$. With $U = 0.4$: $F_{\mathbf{X}}(\mathbf{x}) \approx 0.4$. There are multiple $\mathbf{x}$'s for which $F(\mathbf{x}) = 0.4$.

1.6 Generating Multinomial via Conditional Distributions

For $\mathbf{X} \sim \text{Mult}(n, \mathbf{p})$:

Do n independent trials, each trial can lead to k outcomes. The j -th outcome happens with probability p_j ; we have $n, \mathbf{p}, U$ and the cumulative probabilities $p_1, p_1 + p_2, \dots, \sum_{j=1}^k p_j$.

If $\sum_{j=1}^{i-1} p_j < U < \sum_{j=1}^i p_j$, the outcome of trial 1 is the vector $(0, 0, \dots, \underbrace{1}_{i\text{th}}, 0, \dots, 0)$.

$$\begin{array}{ccccccc} | & p_1 & | & p_2 & | & p_3 & | & \dots & | & p_k & | \\ \hline & & & & & & & & & & \\ 0 & & & & & & & & & & 1 \end{array}$$

Roll a die use n many uniforms. *Inefficient*: requires n many uniforms.

Alternatively, use conditional distributions. $\mathbf{X} \sim \text{Mult}(n, \mathbf{p})$, $\mathbf{x} = (x_1, \dots, x_k)^T$:

- Marginal: $X_1 \sim \text{Bin}(n, p_1)$.
- Conditional: $X_2 \mid X_1 \sim \text{Bin}\left(n - X_1, \frac{p_2}{p_2 + \dots + p_k}\right)$.
- $\begin{pmatrix} X_2 \\ \vdots \\ X_k \end{pmatrix} \mid X_1 \sim \text{Mult}\left(n - X_1, \begin{pmatrix} p_2 \\ \vdots \\ p_k \end{pmatrix} / \sum_{j \geq 2} p_j\right)$.
- $X_3 \mid (X_1, X_2) \sim \text{Bin}\left(n - X_1 - X_2, \frac{p_3}{p_3 + \dots + p_k}\right)$.
- $X_{k-1} \mid \cdot \sim \dots$ and $X_k = n - \sum_{i=1}^{k-1} X_i$.

Generate binomials by inverse transform using $(k - 1)$ uniforms.

1.7 Generating Univariate Normal

$X \sim N(\mu, \sigma^2)$: if $Z \sim N(0, 1)$ then $X = \mu + \sigma Z$.

Enough to develop this for $N(0, 1)$.

1.7.1 Inverse Transform Approximation

$F \rightarrow F^{-1}$; no closed form for F (i.e. Φ).

Approximation:

$$\bar{\Phi}^{-1}(x) = ? \quad \bar{\Phi}(x) \approx 1 - \Phi(x)(b_1 t + b_2 t^2 + \dots + b_5 t^5), \quad t = \frac{1}{1 + px},$$

with max difference 7.5×10^{-8} , $x > 0$. Here $b_1, \dots, b_5, p$ are known constants and $\bar{\Phi}(-x) = 1 - \bar{\Phi}(x)$ for $x < 0$.

Inverse of $\bar{\Phi}^*$: solve $\bar{\Phi}^*(x) - U = 0$ numerically (e.g. Newton–Raphson). R uses this inversion by default. Lots of approximations involved not exact and takes a lot of time.

1.7.2 Box–Muller Method

Generate $N(0, 1)$ in *pairs*. Need two uniforms.

Let $R \sim \text{Exp}(2) \perp\!\!\!\perp \Theta \sim U(0, 2\pi)$, i.e.:

$$\begin{aligned} U_1 &\sim U(0, 1) \perp\!\!\!\perp U_2 \sim U(0, 1) \\ R &= -2 \log U_1, \quad \Theta = 2\pi U_2. \end{aligned}$$

Set

$$X_1 = \sqrt{R} \cos \Theta, \quad X_2 = \sqrt{R} \sin \Theta.$$

Then X_1, X_2 are *independent* $N(0, 1)$.

Verification. The joint density of (R, Θ) :

$$f(R, \Theta) = \frac{1}{2} e^{-R/2} \cdot \frac{1}{2\pi}, \quad R > 0, 0 < \Theta < 2\pi.$$

The Jacobian of the inverse transform $(X_1, X_2) \mapsto (R, \Theta)$:

$$J = \begin{pmatrix} \frac{1}{2\sqrt{R}} \cos \Theta & \frac{1}{2\sqrt{R}} \sin \Theta \\ -\sqrt{R} \sin \Theta & \sqrt{R} \cos \Theta \end{pmatrix}, \quad |\det J| = \frac{1}{2}.$$

$\det$ of inverse transform = 2 Therefore $(X_1, X_2) \in \mathbb{R}^2$ and

$$f(x_1, x_2) = \frac{1}{2} \cdot \frac{1}{2\pi} \cdot 2 e^{-(x_1^2 + x_2^2)/2} = \frac{1}{\sqrt{2\pi}} e^{-x_1^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-x_2^2/2}.$$

1.8 Generating Multivariate Normal

Let $\mathbf{X} \sim N_d(\boldsymbol{\mu}, \Sigma)$ with Σ positive definite.

If $\Sigma = AA^T$ and $\mathbf{Z} \sim N_d(\mathbf{0}, I)$, then

$$\mathbf{X} = \boldsymbol{\mu} + A\mathbf{Z} \sim N(\boldsymbol{\mu}, \Sigma).$$

We know how to generate $N(0, 1)$ and $\mathbf{Z}$ is a vector of d many independent $N(0, 1)$'s.

1.8.1 Cholesky Decomposition

Cholesky decomposition gives A from Σ . Requires $O(d^2/2)$ computations. Efficient. A is triangular.

1.8.2 Eigen-Decomposition / Singular Value Decomposition

$$\Sigma = V\lambda V^T,$$

where λ is the diagonal matrix of eigenvalues of Σ , and V is the matrix whose columns are the eigenvectors.

Set $A = V\lambda^{1/2}$; then $AA^T = \Sigma$.

Principal Components

Note $X = AZ \Rightarrow Z = A^{-1}X = \lambda^{-1/2}V^T X$ (if X is centred, i.e. $X \leftarrow X - \boldsymbol{\mu}$).

Order eigenvalues in **decreasing** order. All components of Z are linear combinations of X :

$$Z = \begin{pmatrix} Z_1 \\ \vdots \\ Z_d \end{pmatrix}, \quad Z_j = \lambda_j^{-1/2} \mathbf{v}_j^T X \quad (d\text{-dimensional}).$$

Z_j are independent of each other.

The variance of $\mathbf{v}_j^T X$ is called the j -th *principal component*:

$$\text{Var}(\mathbf{v}_j^T X) = \lambda_j.$$